

Christopher Anaya

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SUMMARY

Full-stack software engineer building React, Go, Python, and Postgres systems for model-backed workflows. I turn ambiguous AI behavior into bounded product flows with review states, deterministic checks, evaluation gates, source inspection, and artifact-backed reports.

EXPERIENCE

Independent Open-Source Software Engineer | Mar 2026 - Present

- Meal-plan checking needs semantic interpretation without letting the model make nutrition decisions, so I built MealCheck as a React/Vite product over a Go API/worker, Postgres run state, filesystem artifacts, Cloudflare deployment, and private llama.cpp normalization.
- Local-model workflows fail when users cannot see what the model inferred, so I added source-linked review, correction, rejection, rewrite, progress, deletion, report, recommendation, and source-inspection flows before and after deterministic checking.
- AI product claims need repeatable evidence, so I added P0/P1 evaluations, live local-model smoke paths, model-comparison artifacts, Python operator summaries, and Go/TypeScript CI gates that reviewers can inspect.

Graduate Student Researcher, LASIGE / University of Lisbon | Oct 2023 - Jun 2025

- Biomedical QA models can look stable until question wording changes, so I built a fine-tuning and evaluation workflow with fixed seeds, deterministic perturbations, BioASQ-style scoring, and saved clean-vs-perturbed artifacts.
- The thesis workflow needed to be rerunnable after the competition, so I packaged it as BioLLM-Finetune with manifests, tables, figures, validation metadata, and report artifacts that supported a CLEF 2024 workshop publication.

SELECTED PROJECTS

MealCheck

- Users need to know whether MealCheck blocked a plan because of their input, the local model, or deterministic rules; completed reports retain source rows, normalized rows, repairs, unresolved-food summaries, recommendations, and guideline citations.
- To keep the hosted scope honest, the current one-day local-model path is checked by a 3/3 live P0 repeat regimen and a P1 resolver gate that resolves 885 of 900 strict FNDDS-grounded items with zero expected-outcome mismatches.

LLM Extraction Platform

- Natural-language extraction needs service semantics beyond prompt calls, so I built explicit API contracts, schema-constrained sync/async extraction, runtime policy gates, separate worker execution, request-trace artifacts, and Kubernetes deployment artifacts.
- Reviewers need a short path from claim to evidence, so the repo includes five canonical proof points, CI-backed tests, and evidence manifests that tie runtime behavior to inspectable artifacts.

TECHNICAL STRENGTHS

Languages Go, TypeScript/React, Python, SQL/Postgres, HTML, CSS

Software & Tools REST APIs, Vite, JSON Schema, llama.cpp, FastAPI, Docker, Kubernetes, Cloudflare, Redis, Prometheus/Grafana, CI/CD, Git

EDUCATION

- MSc, Data Science | Faculty of Science, University of Lisbon (2022 - 2025)
- MSc, Mathematical Research | University of Valencia (2015 - 2016)
- BA, Astronomy (Minor: Applied Mathematics) | University of Colorado Boulder (2008 - 2013)

PUBLICATIONS

- Anaya, C.; Fernandes, M.; Couto, F.M. *LLM Fine-Tuning With Biomedical Open-Source Data*. CLEF 2024 (BioASQ 12b).